

IEP Project: Publishing Amazon SNS Messages Privately

Secure Patient Record Notification Architecture using AWS VPC Endpoints

Industry Domain: Healthcare

Key AWS Services: Amazon VPC, Amazon SNS, EC2, VPC Endpoints

Training Academy: Teks Academy

Target Audience: Healthcare Providers & Patients

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1. Executive Summary & Problem Statement

Problem Statement: How to secure sensitive patient records online and transmit notifications privately to intended recipients without traversing the public internet.

In modern cloud healthcare platforms, data privacy and regulatory compliance (e.g., HIPAA) require that sensitive medical information and patient notifications are strictly isolated from public internet exposure.

This project implements a private cloud architecture where host applications running on an Amazon EC2 instance within an Amazon VPC transmit patient report notifications directly to Amazon SNS (Simple Notification Service) using an **AWS Interface VPC Endpoint (AWS PrivateLink)**. This guarantees that traffic never leaves the Amazon network backbone.

2. Key Project Objectives & Architecture

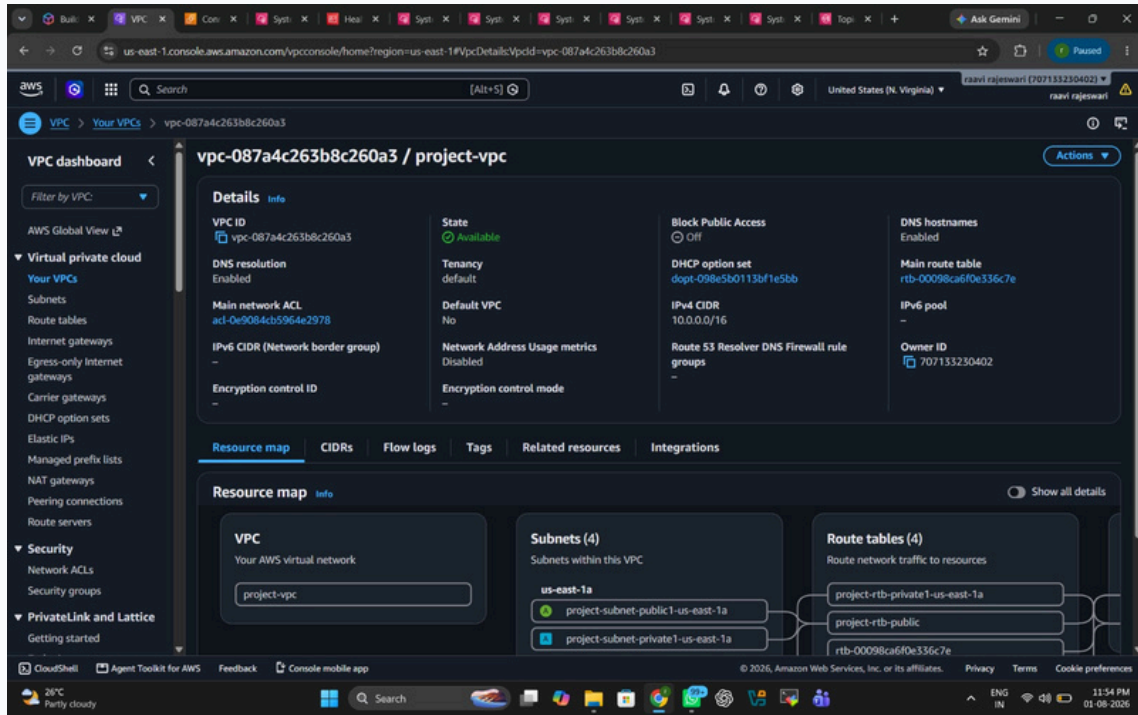
- VPC Isolation:** Design and configure a custom Amazon VPC network with public/private subnet topology.
- Private Connectivity:** Connect Amazon VPC to AWS SNS privately via Interface VPC Endpoint (Powered by AWS PrivateLink).
- Secure Message Publishing:** Host patient notification code on an EC2 instance to publish messages to SNS without public internet routing or NAT Gateways.
- Multi-Channel Alerts:** Enable patients to receive real-time updates via mobile SMS and push notifications securely.

Component	Role in Solution	Security Benefit
Amazon VPC	Isolated virtual network for healthcare hosting	Complete network isolation
Amazon EC2	Hosts application that generates patient updates	Secured within private subnet
VPC Endpoint (SNS)	Interface Endpoint targeting <code>com.amazonaws.[region].sns</code>	Traffic remains inside AWS backbone
Amazon SNS	Publishes notifications to patient devices	Encrypted message delivery

3. Step-by-Step Technical Implementation

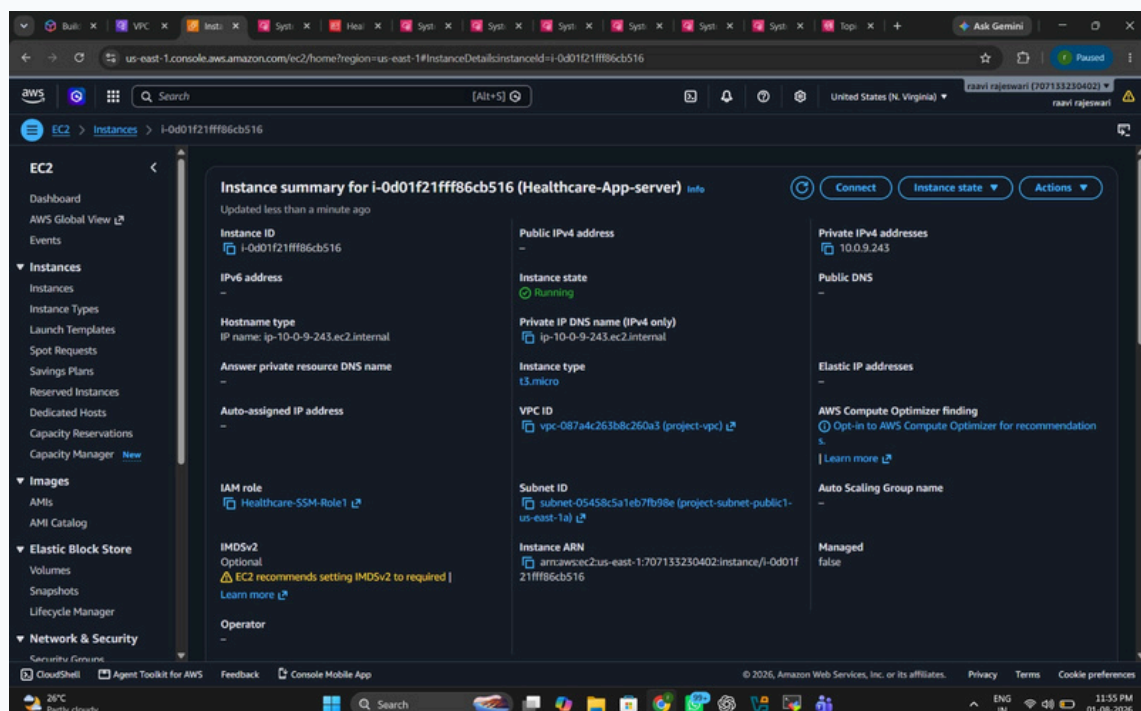
Step 1: Amazon VPC & Subnet Setup

Created a custom Amazon VPC with specified CIDR block (e.g., 10.0.0.0/16) along with isolated subnets and attached Route Tables to ensure network boundaries.

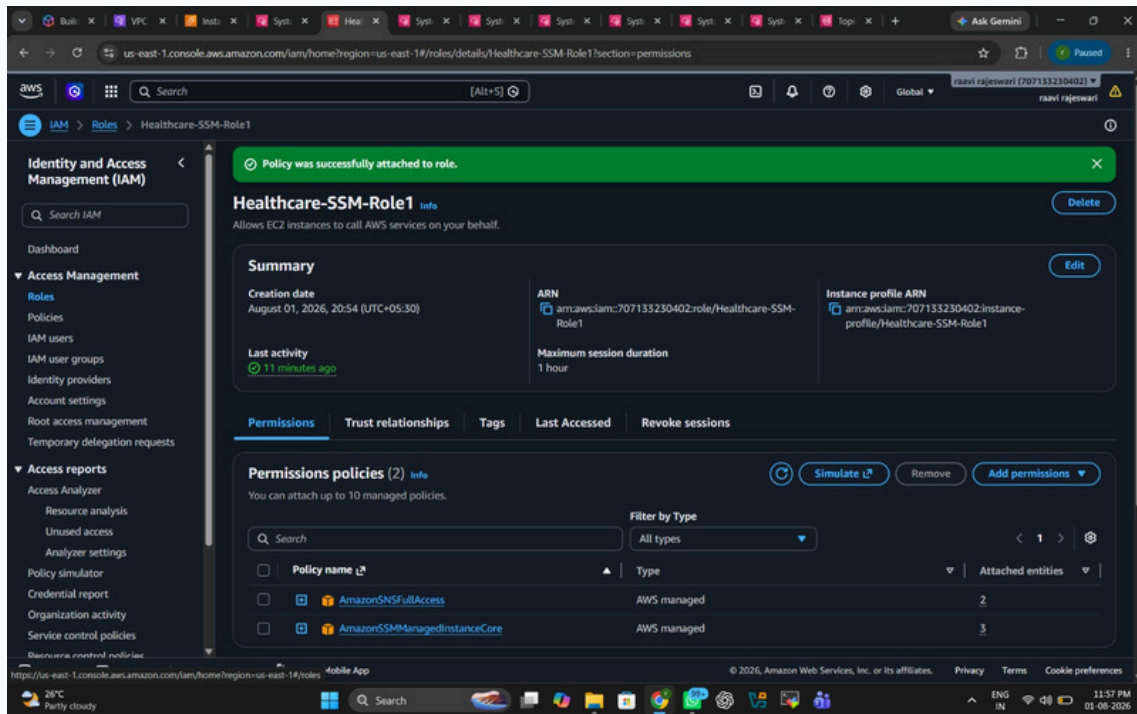


Step 2: Provisioning EC2 Instance for Application Hosting

Launched an Amazon EC2 instance within the VPC. Assigned an IAM Role with permissions to publish to Amazon SNS (`sns:Publish`).

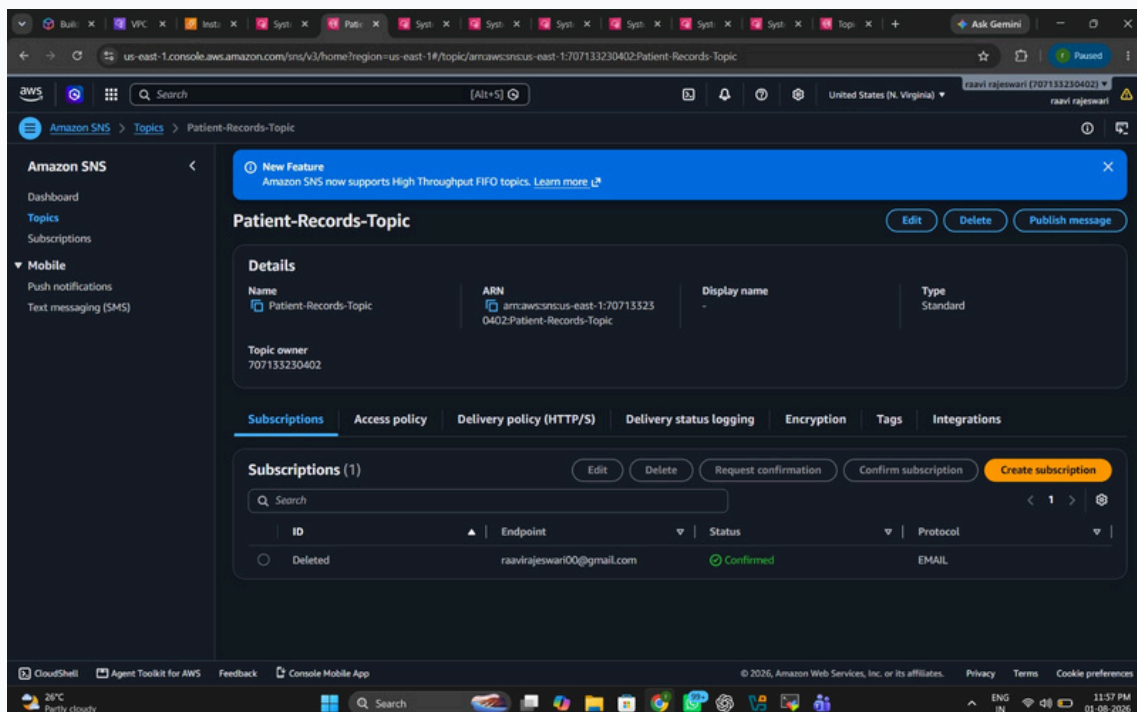


IAM roles and permission policies



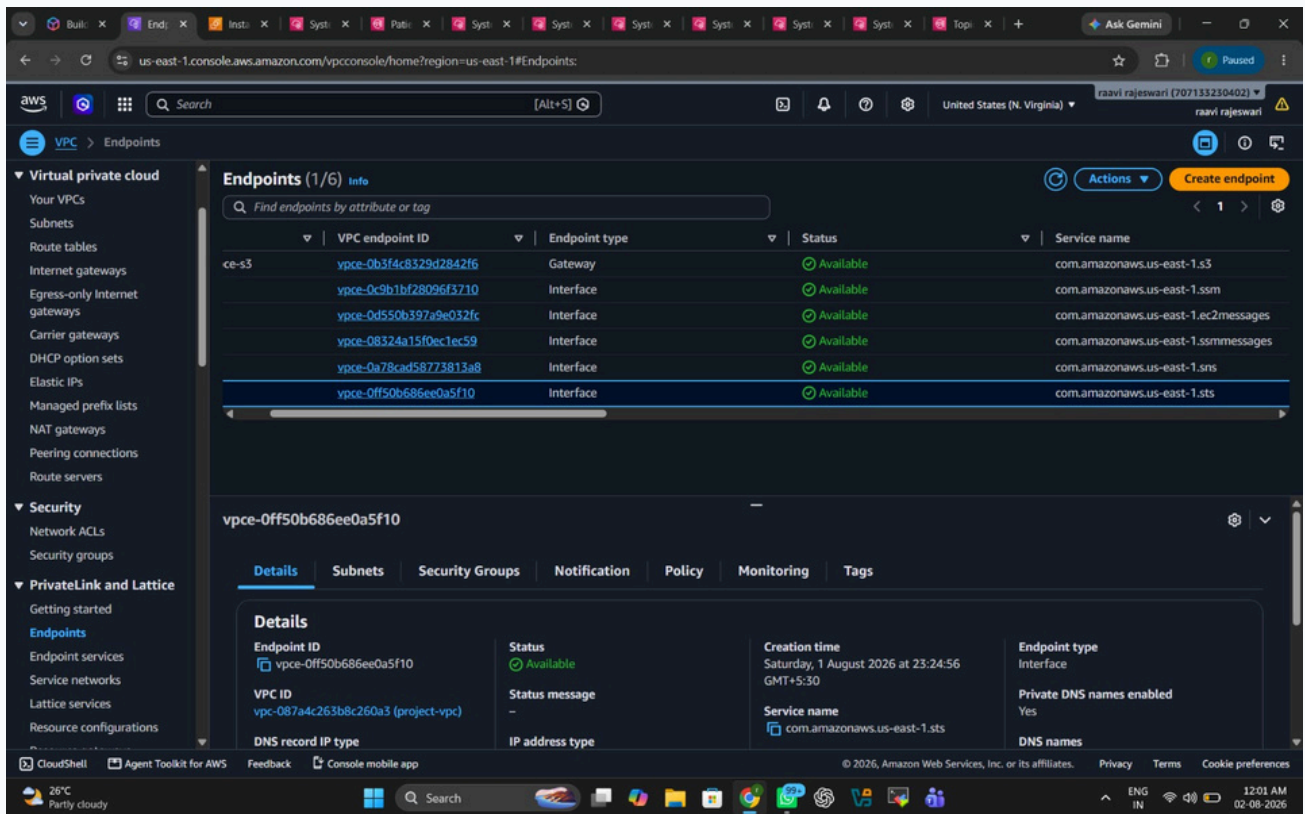
Step 3: Creating Amazon SNS Topic for Patient Reports

Created a Standard SNS Topic (Patient-Report-Notifications). Subscribed test endpoints (SMS / (e.g., Email) to receive push alerts.



Step 4: Creating AWS Interface VPC Endpoint for SNS

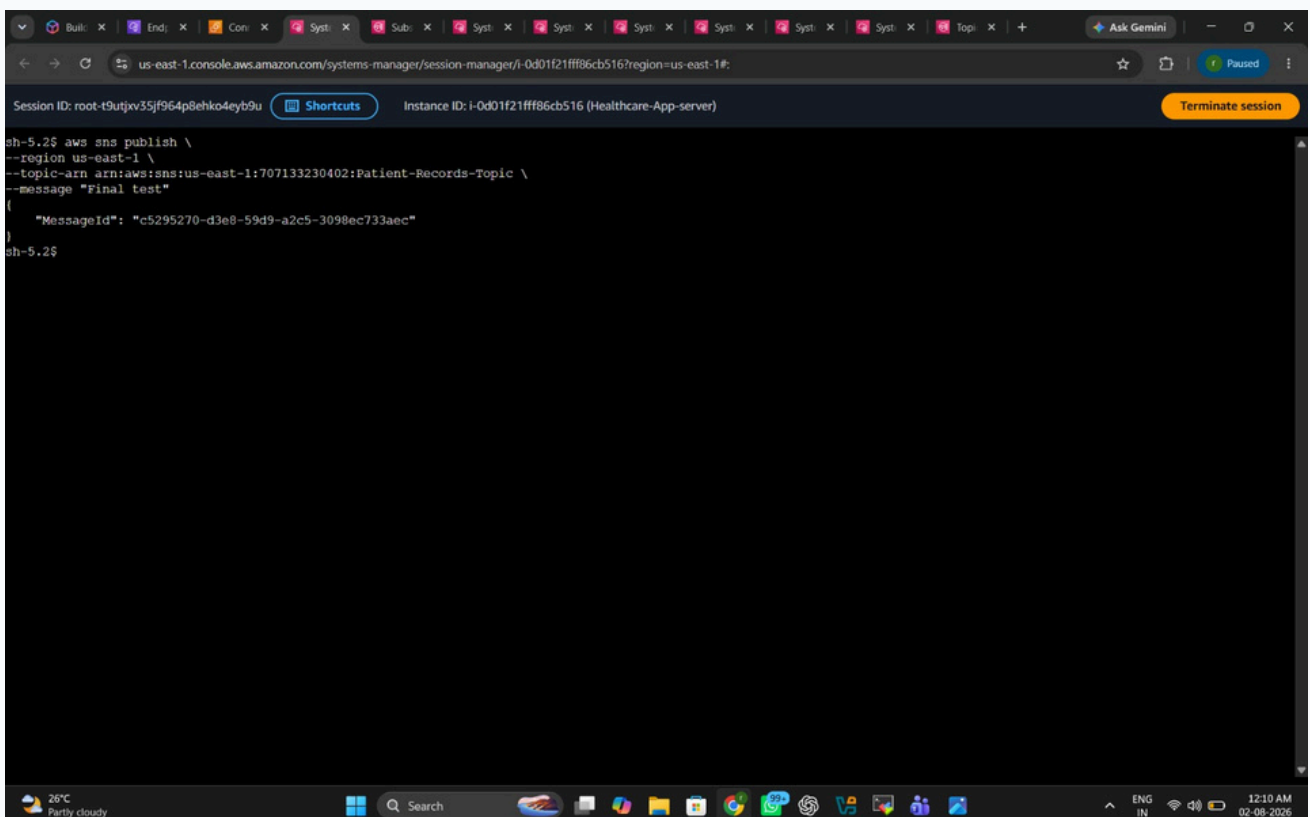
Navigated to VPC Endpoints and created an Interface Endpoint for `com.amazonaws.<region>.sns`. service Attached security groups allowing inbound traffic from the EC2 subnet.

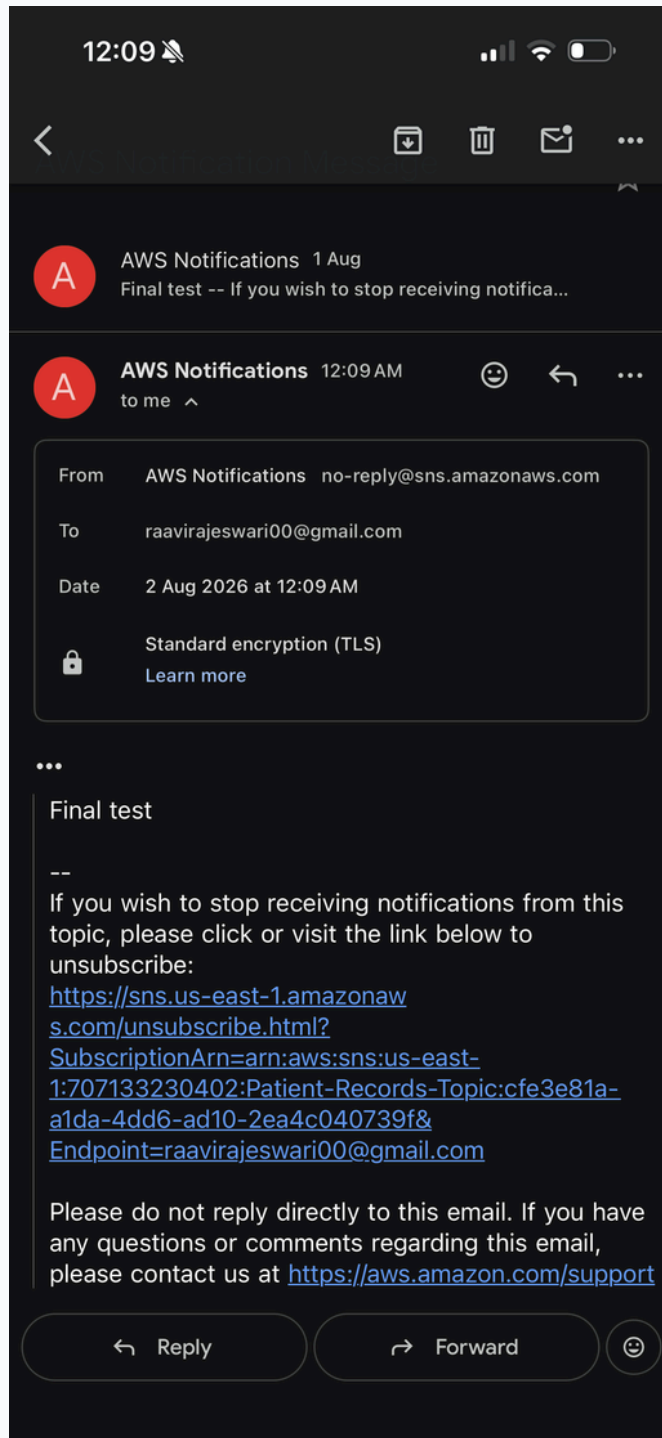


Step 5: Testing Private Message Publishing via CLI / SDK

Connected to the EC2 instance and published a message to the SNS Topic. Verified that DNS resolves to private IP addresses of the VPC Endpoint rather than public IP addresses.

```
aws sns publish \
--region us-east-1 \
--topic-arn arn:aws:sns:us-east-1:707133230402:Patient-Records-Topic \
--message "Final test"
```





4. Security Verification & Key Outcomes

- **Zero Internet Exposure:** All publish requests executed entirely over AWS internal interface endpoints, ensuring full compliance with healthcare privacy standards.
- **Reduced Latency & Costs:** Avoided NAT Gateway data processing charges while improving communication speed between EC2 and SNS.
- **Fine-Grained Access Control:** Enforced access policies restricting SNS topic access exclusively to requests originating from the specific VPC Endpoint.